



Anglo-Chinese School (Independent)
PRELIMINARY EXAMINATION
YEAR FOUR EXPRESS

BIOLOGY

6093/01

Paper 1 Multiple Choice

15 August 2024, Thursday

Additional Materials: Multiple Choice Answer Sheet

1 hour

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, highlighters, glue or correction fluid/tape.

Write your name and index number on the Answer Sheet in the spaces provided.

There are **forty** questions on this paper. Answer **all** questions. For each question, there are four possible answers, **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

The use of an approved scientific calculator is expected, where appropriate.

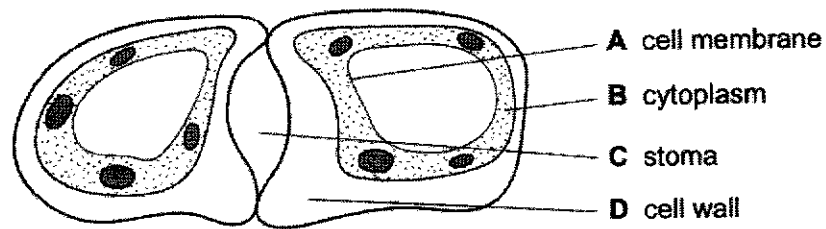
This document consists of **18** printed pages, including this cover page.

6093/01/Y4Exp/Prelim/Aug2024

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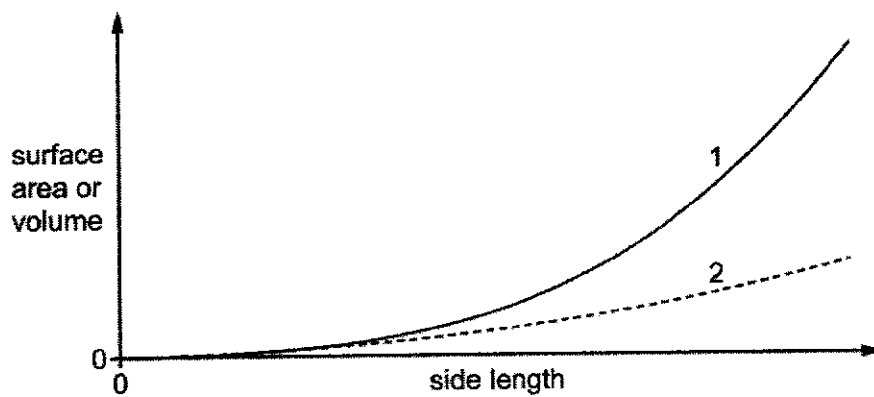
- 1 The diagram shows a student's drawing of guard cells.

Which labelling line is **incorrect**?



Correct Answer : A

- 2 The graph shows the effect of increasing the side length of agar cubes on the surface area and the volume of the cubes.



Which correctly identifies lines 1, 2 and the effect of increasing side length on the surface area to volume ratio of the cubes?

	line 1	line 2	surface area to volume ratio
A	surface area	volume	decreases
B	surface area	volume	increases
C	volume	surface area	decreases
D	volume	surface area	increases

Correct Answer : C

- 3 The table shows the relative concentrations of two mineral ions in a root cell and in the water in the soil around the plant root.

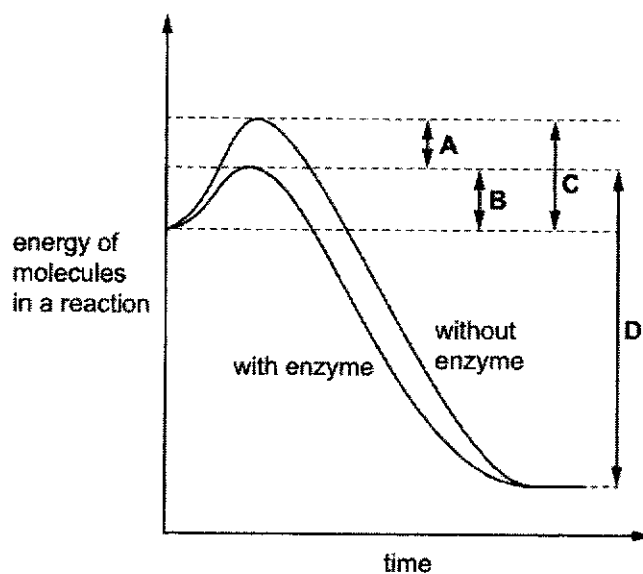
	concentration of magnesium ions / arbitrary units	concentration of nitrate ions / arbitrary units
water around root	23	19
inside root cell	2475	1306

What is needed for the plant root to increase the concentration of the mineral ions in its cells?

- A energy for active transport
- B energy for osmosis
- C water for active transport
- D water for osmosis

Correct Answer : A

- 4 Which region on the graph shows the activation energy of an enzyme-catalysed reaction?

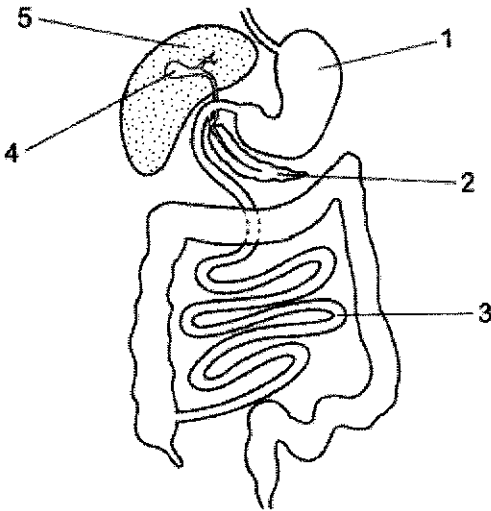


- 5 Which statement about chemical digestion in the human alimentary canal is correct?

- A Digestion of carbohydrates is completed in the colon.
- B Enzymes are secreted to break down cellulose in the duodenum.
- C Protein digestion is completed in the ileum.
- D The stomach secretes enzymes to break down starch.

Correct Answer : B

6 The diagram shows part of the human alimentary canal and associated organs.



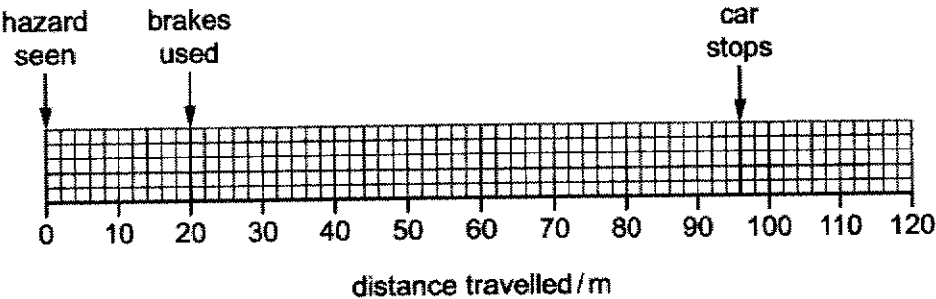
Which row describes the functions of parts shown in the diagram?

	Structure	Function	Structure	Function
A	1	Digestion of protein	3	Absorption of products from digestion
B	2	Emulsifying of fats	3	Absorption of amino acids and glucose
C	4	Production of bile	5	Making digestive enzymes
D	4	Storing digestive enzymes	2	Making digestive enzymes

Correct Answer: A

7 The chart gives information about a car being driven at 110 km/h by someone who has **not** consumed alcohol.

It shows how far the car travels from the time the driver sees a hazard and reacts by using the brakes and how far the car will then travel before finally stopping.



If the same driver consumes alcohol before driving, the distance travelled before reacting to the hazard increases by 20%.

Correct Answer : B

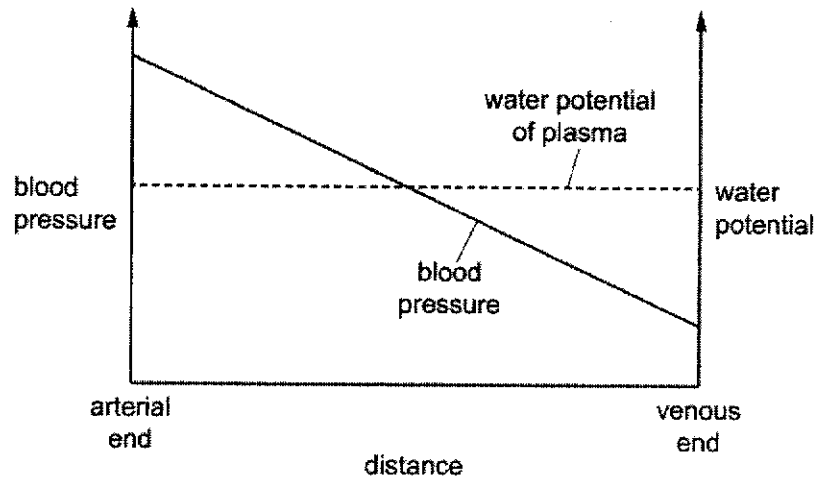
What would then be the total distance travelled by the car before stopping?

- A 96 m B 100 m C 111.2 m D 115.2 m

Correct Answer: B

5

- 8 The graph shows the changes in blood pressure and water potential across a blood capillary from the arterial end of the capillary to the venous end of the capillary.



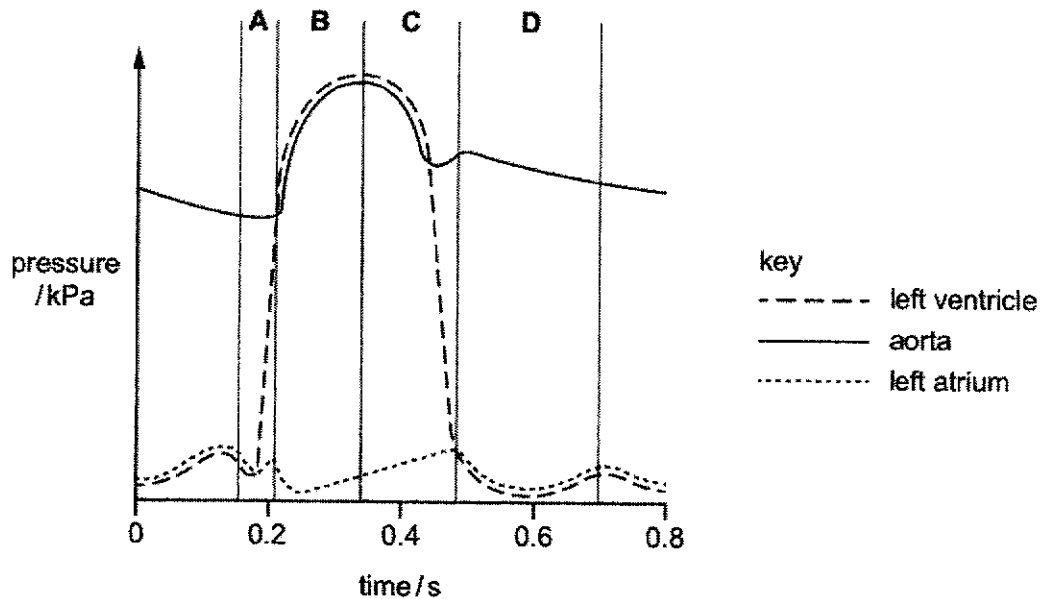
What could result in an accumulation of excess tissue fluid?

- A an increase in concentration of large proteins in the blood plasma
- B an increase in blood pressure
- C a decrease in the concentration of small proteins in tissue fluid
- D a decrease in blood pressure

Correct Answer: B

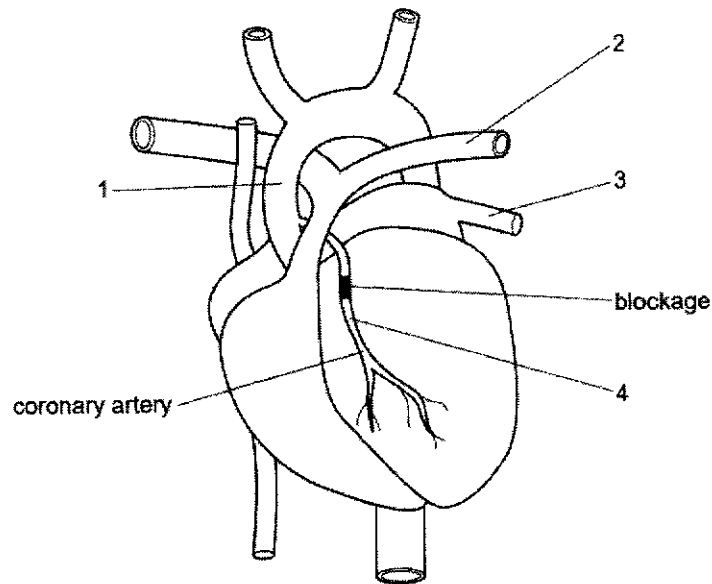
- 9 The diagram shows the pressure changes in various structures of the left side of the heart during the cardiac cycle.

At the end of which period is the ventricle full of blood?



Correct Answer: A

- 10 The diagram shows an external view of the heart of a patient with a blockage of the coronary artery. This could be treated by inserting a tube to bypass the blockage.



Which two vessels would be joined by this tube?

- A** 1 and 2 **B** 1 and 4 **C** 2 and 4 **D** 3 and 4

Correct Answer: B

- 11 Which row correctly identifies the effects of carbon monoxide, nicotine and tar?

	effect		
	raises blood pressure	causes gene mutation	reduces oxygenation of blood
A	carbon monoxide	nicotine	tar
B	nicotine	nicotine	carbon monoxide
C	nicotine	tar	carbon monoxide
D	tar	carbon monoxide	nicotine

Correct Answer: C